

BASiS: Adapting Conversational Behavior in Purpose-Specific Dialogue Systems through Bayesian Data Selection

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This paper proposes BASiS (Bayesian Style-based Selection), a method for adapting a local large language model (LLM) to the conversational style required for purpose-specific dialogue through Bayesian data selection guided by a small, high-quality corpus. Recent LLMs can produce fluent dialogue and accurate responses. In purpose-specific settings such as emotional support and medical history-taking, however, conversational style—*how* the system responds, not only *what* it says—is also crucial. Although small, high-quality dialogue corpora capture styles suited to particular purposes, they contain too little data to train an LLM directly. Large general-purpose dialogue corpora provide sufficient volume, but they do not necessarily contain enough dialogues that match the target style.

BASiS extracts conversational states, response strategies, and state transitions from a small, high-quality corpus and represents them as a Bayesian dialogue model. It then uses this model to assess dialogue data in a large general-purpose corpus according to their compatibility with the target conversational style and selects data for training. Finally, BASiS constructs preference pairs from the selected data and trains a local LLM using Direct Preference Optimization (DPO). This formulation does not require detailed, purpose-specific guidelines to be written manually and represents, as a probability distribution, the conversational diversity in which multiple response strategies may be valid for the same context.

We evaluated BASiS in three types of purpose-specific dialogue: emotional support, mathematics tutoring, and medical history-taking. In each setting, we compared an untrained base model, a model trained on data selected by BASiS, and a model trained on randomly selected data. We evaluated how well each model reproduced the characteristic response policies and dialogue progression of the target corpus and analyzed how training affected general conversational quality. Across all three experimental settings, the model trained on BASiS-selected data reproduced the characteristic conversational style of the target corpus to the greatest extent among the three conditions.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; **Discourse, dialogue and pragmatics**; Learning from implicit feedback; • **Human-centered computing** → *Natural language interfaces*.

Additional Key Words and Phrases: purpose-specific dialogue systems, conversational style, data selection, Bayesian dialogue modeling, preference optimization, emotional support dialogue, large language models

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1 Introduction

Advances in instruction tuning and reinforcement learning from human feedback (RLHF) have enabled recent large language models (LLMs) to follow instructions, generate accurate responses, and sustain natural and coherent multi-turn conversations across a wide range of topics [11]. In purpose-specific dialogue such as emotional support, mathematics tutoring, and medical history-taking, however, a system must do more than provide correct information: it must choose

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an appropriate response policy based on the interlocutor’s state and the current stage of the dialogue [7, 13]. For example, it may need to acknowledge a person’s feelings before offering advice, encourage a learner to reason independently, or elicit necessary symptoms in an appropriate order. Such conversational styles are reflected in small, high-quality corpora developed for specific purposes, but these corpora are limited in size for direct use in LLM training. Large general-purpose corpora offer greater volume and diversity, yet they do not necessarily include enough dialogues that match the target style. A method is therefore needed to use the style captured in a small, high-quality corpus to select suitable dialogue data from a large general-purpose corpus.

Several approaches have been explored to generate responses suited to purpose-specific dialogue. Guideline-based response control explicitly defines a desirable dialogue policy in advance and guides response generation to follow it. GuidedTOD, proposed by Wen et al., probabilistically represents the flow of task-oriented dialogue and reranks candidate responses while updating a guideline-based policy as the dialogue progresses; it achieved higher guideline compliance than previous methods [16]. Another line of work predicts a response strategy from the current user state. EmoDynamix, proposed by Wan et al., models the relationships between multiple user emotion states and response strategies as a heterogeneous graph, achieving strong strategy-prediction performance while making the selection process interpretable [15]. Research on selecting LLM training data has further shown that carefully chosen, high-quality data can efficiently adapt model behavior even when the selected dataset is small [18, 21].

These approaches nevertheless leave three challenges. First, guideline-based response control requires desirable dialogue policies to be written manually for each purpose or domain. Conversational style is often expressed implicitly through tone, response timing, and progression across multiple turns, making it difficult to describe comprehensively as explicit rules. Second, response-strategy prediction generally predicts a single next strategy for the current state. In real dialogue, however, several strategies may be valid in the same context, such as continuing to empathize, asking a question to explore the issue, or organizing the available information. A single prediction cannot adequately represent the diversity of responses found in a target corpus. Third, many existing data-selection methods focus on general qualities such as accuracy, usefulness, diversity, and task coverage. Even when two responses are comparable on these dimensions, they may differ in when they are given and in how they change the conversational state. Unless dialogue history and transitions between response strategies are included in the selection criterion, it is difficult to reliably select data that preserve the characteristic dialogue flow of the target corpus.

To address these challenges, we propose BASiS (Bayesian Style-based Selection), a method that adapts a local LLM to the conversational style required for purpose-specific dialogue through Bayesian data selection based on a small, high-quality corpus. BASiS first uses an LLM to analyze the small corpus and structures each dialogue in terms of conversational states, which describe the interlocutor’s situation or the stage of the dialogue; response strategies, which describe response policies such as empathy, questioning, and advice; and state transitions caused by the responses. It then constructs a Bayesian dialogue model from this structure, using transition probabilities between states and observation likelihoods for response strategies in each state. The model scores each dialogue instance in a large general-purpose corpus according to its compatibility with the target conversational style and selects data for training. Finally, BASiS constructs preference pairs from the selected data and applies Direct Preference Optimization (DPO) with Low-Rank Adaptation (LoRA) to a local LLM, thereby adapting the model to the style captured in the small corpus.

A central feature of BASiS is that it represents conversational style not as individual response expressions or detailed, manually authored guidelines, but as probabilistic transitions between conversational states and response strategies. This allows the implicit dialogue flow in the corpus to guide data selection without requiring detailed guidelines for each purpose. Rather than treating a single response strategy as the only correct choice, BASiS maintains a probability

distribution over conversational states and evaluates candidate responses through Bayesian updating. It can therefore accommodate conversational diversity in which multiple strategies may be valid for the same context. Moreover, the same procedure can be applied to different purpose-specific settings, including emotional support, mathematics tutoring, and medical history-taking, by changing the small reference corpus and the large corpus from which data are selected. Using the selected data for LoRA-based DPO allows the target style to be incorporated while retaining the general dialogue capabilities of the local LLM.

The remainder of this paper is organized as follows. Section 2 reviews related work on guideline-based response control, dialogue-state-based response-strategy prediction, and training-data selection for LLMs. Section 3 describes the overall architecture of BASiS and the role of each module. Section 4 presents the experiments used to evaluate BASiS, their results, and our discussion. Finally, Section 5 concludes the paper.

2 Related Work

One approach to realizing purpose-specific behavior in dialogue systems is to provide desirable response policies as natural-language guidelines and control response generation accordingly. Gupta et al. proposed DialGuide, which describes in natural language both the conversational contexts in which a guideline applies and the content that a response should include [4]. DialGuide addresses guideline selection, guideline-grounded response generation, and verification that generated responses satisfy the selected guideline. Experiments on open-domain and safety-related dialogue data showed that it can generate safe and engaging responses that follow developer-provided guidelines. Wen et al. introduced a Chained Prior that represents a sequence of task-oriented dialogue guidelines as a Markov chain and proposed GuidedTOD, which updates its policy as the dialogue progresses [16]. By reranking candidate responses with the Chained Prior, GuidedTOD improved guideline compliance over previous methods and increased action-prediction accuracy by approximately 20%. These studies demonstrate the value of expressing desired behavior as explicit guidelines and incorporating them into response selection or generation.

Other studies predict the next response strategy from the user’s state and dialogue history instead of directly applying manually defined guidelines. Cheng et al. proposed MultiESC, which captures fine-grained emotional expressions and their causes and selects a response strategy while anticipating future user reactions [2]. MultiESC uses an A*-inspired lookahead evaluation to select strategies predicted to provide greater long-term support, outperforming previous methods in both strategy planning and response generation. Deng et al. introduced utterance labels and evaluation measures that represent whether the user or the system leads the conversation, and proposed the Knowledge-enhanced Mixed-Initiative model (KEMI), which combines strategy prediction, external-knowledge selection, and response generation [3]. KEMI uses a knowledge graph constructed from mental-health conversations to select strategies and knowledge appropriate to the dialogue context, and it outperformed previous methods in both content preservation and mixed-initiative behavior. Wan et al. proposed EmoDynamix, which models the relationship between fine-grained user emotion states and system response strategies as a heterogeneous graph [15]. EmoDynamix separates strategy prediction from response generation and predicts the next response strategy from relationships in the graph. Experiments on two emotional-support dialogue datasets showed higher strategy-prediction performance than previous methods, reduced bias toward particular strategies, and enabled the strategy-selection process to be traced.

Research on efficient LLM adaptation has also emphasized the quality and utility of training data rather than only its volume. Zhou et al. proposed LIMA, which trains LLaMA on only 1,000 carefully selected instruction–response examples, and showed that even a small amount of high-quality data can teach a model desirable response formats and conversational behavior [21]. In human evaluation, LIMA responses were judged at least as good as GPT-4 responses in

43% of cases and performed strongly against several models trained with much larger alignment datasets. Chen et al. proposed AlpaGasus, which uses a strong LLM to assess each instruction–response example and remove low-quality data [1]. AlpaGasus was trained on only 9,000 of Alpaca’s 52,000 examples, yet it outperformed Alpaca trained on the full dataset in both automatic and human evaluation while substantially reducing training time. Liu et al. proposed DEITA, which separately evaluates the complexity, quality, and diversity of instruction data and combines these criteria for selection [8]. Using 6,000 supervised fine-tuning examples—less than one-tenth the amount used by its baselines—DEITA achieved performance comparable to or better than leading open-source models at the time. Xia et al. proposed LESS, which estimates how strongly each training example affects acquisition of a target capability from the gradient similarity between that example and a small set of target-task examples [17]. Training on only 5% of the selection pool could outperform training on the full dataset across several downstream tasks, demonstrating the effectiveness of purpose-driven data selection.

These studies have advanced response-policy control, response-strategy prediction, and efficient selection of training data, but three major challenges remain from the perspective of conversational-style selection. First, guideline-based methods such as DialGuide and GuidedTOD require desirable dialogue policies to be written manually for each purpose or domain. Features expressed implicitly in a corpus—including tone, response timing, and progression across multiple turns—are difficult to specify comprehensively as guidelines, making their design costly. Second, response-strategy prediction methods such as MultiESC, KEMI, and EmoDynamix can choose strategies based on the user state and anticipated future dialogue, but ultimately select a particular strategy at each turn and generate a response from it. Because multiple strategies may be equally valid in the same context, a single prediction cannot fully represent the response diversity of a target corpus. Third, LIMA demonstrates the effectiveness of training on small, high-quality datasets, while AlpaGasus and DEITA select data according to response quality, complexity, and diversity, and LESS selects data according to their influence on a small set of target-task examples. Even when responses are comparable in general quality or utility for the target task, however, their compatibility with the target style may differ because of their timing or the conversational-state transitions they induce. A selection criterion that does not incorporate dialogue history, response strategy, and state transitions cannot adequately distinguish these differences or reliably preserve the characteristic dialogue flow of the target corpus.

3 BASiS

3.1 Concept

We propose BASiS (Bayesian Style-based Selection), which adapts a local large language model (LLM) to the conversational style required for purpose-specific dialogue through Bayesian data selection based on a small, high-quality corpus. BASiS converts the response policies and dialogue progression expressed implicitly in the small corpus into a form suitable for training a local LLM. Rather than training directly on the small corpus, it extracts conversational states, response strategies, and state transitions, constructs a probabilistic dialogue model, and uses that model as a criterion for selecting data from a large general-purpose corpus.

BASiS evaluates each dialogue instance in the large general-purpose corpus according to its compatibility with the conversational style characteristic of the target corpus. It incorporates the dialogue flow in the small corpus into the training data without requiring detailed, manually written dialogue guidelines for each purpose or domain. By maintaining the conversational state as a probability distribution rather than committing to a single state, BASiS

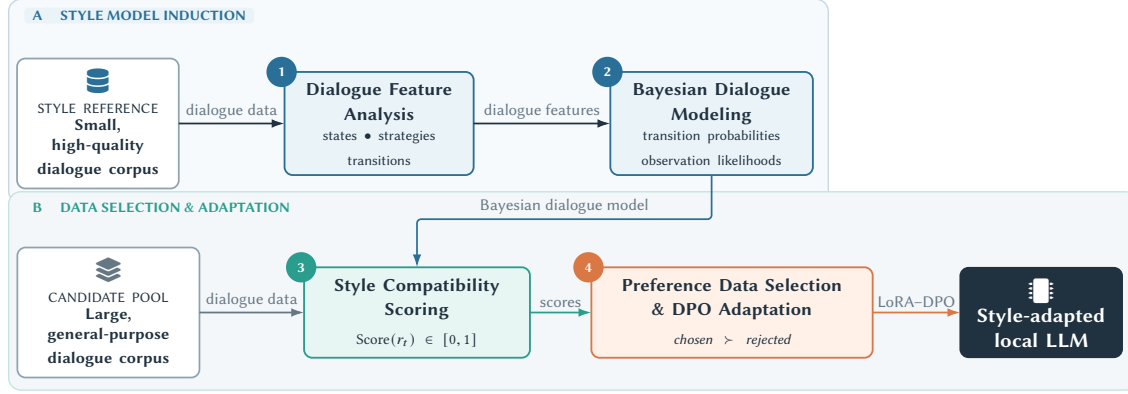


Fig. 1. Overall architecture of BASiS. The upper pathway extracts conversational features from a small, high-quality corpus and constructs a Bayesian dialogue model. The lower pathway uses the model to score candidate responses from a large general-purpose corpus and adapts a local LLM through LoRA–DPO using the resulting preference data. The numbered circles correspond to the four modules described in the text.

represents settings in which multiple response strategies may be valid for the same context. The selected data are ultimately used for DPO training, which adapts the local LLM to the conversational style of the small corpus.

3.2 BASiS Architecture

Figure 1 shows the overall architecture of BASiS. BASiS consists of four modules: the **Dialogue Feature Analysis Module**, **Bayesian Dialogue Modeling Module**, **Style Compatibility Scoring Module**, and **Preference Data Selection and DPO Adaptation Module**.

First, the Dialogue Feature Analysis Module analyzes dialogues in the small, high-quality corpus and extracts conversational states, response strategies, and state transitions as conversational features. The Bayesian Dialogue Modeling Module then aggregates the occurrence patterns of these features and constructs a Bayesian dialogue model consisting of state-transition probabilities and observation likelihoods. The Style Compatibility Scoring Module uses this model to score each candidate response in the large general-purpose corpus according to its compatibility with the target conversational style. Finally, the Preference Data Selection and DPO Adaptation Module selects training data based on these scores, constructs preference pairs consisting of *chosen* and *rejected* responses, and trains a local LLM with LoRA-based DPO.

In this architecture, the small, high-quality corpus serves as the reference from which the conversational style is extracted, whereas the large general-purpose corpus serves as the candidate pool from which training data are selected. The same pipeline can be applied to different types of purpose-specific dialogue by changing the small reference corpus and the large selection corpus.

Dialogue Feature Analysis Module

The Dialogue Feature Analysis Module uses an LLM to analyze each dialogue in the small, high-quality corpus and structures the dialogue progression characteristic of the target corpus. It represents each turn in terms of three elements: a *conversational state*, a *response strategy*, and a *state transition*.

The conversational state s_t represents the interlocutor’s situation or the stage of the dialogue at turn t . Examples include disclosing emotions or considering solutions in emotional support, attempting a line of reasoning or organizing

an error in mathematics tutoring, and presenting symptoms or confirming additional information in medical history-taking. The observation label o_t , which represents the response strategy, describes the policy taken by the respondent in that state; examples include empathy, questioning, information organization, hints, information provision, and advice. The state transition describes how the conversational state changes from s_{t-1} to s_t as a result of choosing response strategy o_t .

Applying this analysis to the entire small corpus yields a collection of tuples (s_{t-1}, o_t, s_t) . These tuples capture not only surface vocabulary and individual responses, but also the dialogue structure: which strategies tend to be used in particular states and which states tend to follow. This structure converts the conversational style implicit in the corpus into a form that can be used by the subsequent modules.

Bayesian Dialogue Modeling Module

The Bayesian Dialogue Modeling Module constructs a Bayesian dialogue model of the dialogue flow in the target corpus from the features produced by the Dialogue Feature Analysis Module. Let $S = \{s_1, \dots, s_K\}$ denote the set of conversational states and $O = \{o_1, \dots, o_L\}$ the set of observation labels corresponding to response strategies.

The Bayesian dialogue model consists of the state-transition probability $P(s_t | s_{t-1})$ and the observation likelihood $P(o_t | s_t)$. The transition probability represents the probability of moving from conversational state s_{t-1} to s_t in the target corpus. Dialogue flows frequently observed in the target corpus receive high probabilities, whereas rarely observed flows receive low probabilities. The observation likelihood represents the probability that response strategy o_t is observed when the dialogue is in state s_t .

Together, these distributions represent which conversational states are likely to occur in the target corpus and which response strategies are likely to be used in each state. The model does not merely memorize utterances found in the small corpus; instead, it represents probabilistic relationships between states and strategies. It can therefore evaluate the compatibility of new candidate responses that do not appear in the small corpus with the target conversational style.

Style Compatibility Scoring Module

The Style Compatibility Scoring Module uses the Bayesian dialogue model to evaluate a candidate response r_t from the large general-purpose corpus according to its compatibility with the target conversational style. Figure 2 illustrates this process.

Before a candidate response is presented, the dialogue state is maintained as a probability distribution over the state set S rather than fixed to a single state. Given the state distribution through turn $t - 1$, denoted by $b_{t-1}(s)$, the predicted state distribution for the next turn is computed using the transition probabilities:

$$\hat{b}_t(s_t) = \sum_{s_{t-1} \in S} P(s_t | s_{t-1}) b_{t-1}(s_{t-1}). \quad (1)$$

Next, an LLM classifies candidate response r_t into observation label o_t using the same label scheme as the Dialogue Feature Analysis Module. The predicted distribution $\hat{b}_t(s_t)$ is updated according to Bayes' rule using this observation label and the observation likelihood:

$$b_t(s_t) = \frac{P(o_t | s_t) \hat{b}_t(s_t)}{\sum_{s' \in S} P(o_t | s') \hat{b}_t(s')}. \quad (2)$$

Here, $b_t(s_t)$ is the posterior probability $P(s_t | o_{1:t})$ conditioned on the observation history $o_{1:t}$ after incorporating the observations into the predicted distribution $\hat{b}_t(s_t)$.

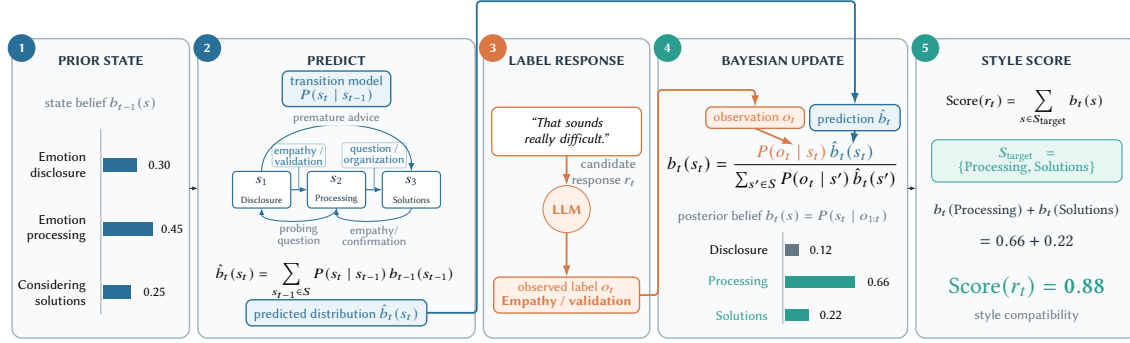


Fig. 2. Schematic example of candidate-response evaluation in the Style Compatibility Scoring Module. The state distribution through the previous turn is predicted using the transition model and then updated with the observation label derived from the candidate response. The blue path indicates where the predicted distribution enters the update equation, and the orange path indicates where the observation label enters it. In this example, the probabilities of the target states Processing and Solutions in the updated distribution are summed to obtain $\text{Score} = 0.88$.

Finally, let $S_{\text{target}} \subseteq S$ be the set of states defined as desirable for the target conversational style. The compatibility score of candidate response r_t is the total probability assigned to S_{target} in the updated state distribution:

$$\text{Score}(r_t) = \sum_{s \in S_{\text{target}}} b_t(s). \quad (3)$$

The compatibility score ranges from 0 to 1. A larger value indicates a higher probability that the candidate response will move the dialogue toward states characteristic of the target corpus. Depending on the domain, S_{target} may include states such as processing emotions or considering appropriate solutions in emotional support, advancing the learner’s own reasoning in mathematics tutoring, or progressively organizing necessary symptom information in medical history-taking.

This module evaluates candidate responses while maintaining a probability distribution over states instead of fixing a single response strategy as the correct next action. Even when each candidate response receives one observation label, responses associated with different strategies can each obtain a high compatibility score if those strategies have high observation likelihoods in the desirable states. Thus, when several response strategies are valid for the same context, the module can evaluate them without restricting the choice to a single strategy.

Preference Data Selection and DPO Adaptation Module

The Preference Data Selection and DPO Adaptation Module selects responses for training from the large general-purpose corpus based on the compatibility scores produced by the Style Compatibility Scoring Module. Among the candidate responses constructed for each dialogue context, a response with high compatibility with the target style is selected as *chosen*, while a response with low compatibility is selected as *rejected*, forming a preference pair.

The resulting preference pairs are used to train the local LLM through Direct Preference Optimization (DPO) [12]. DPO directly optimizes the model so that chosen responses are preferred to rejected responses. Rather than updating all model parameters, we apply Low-Rank Adaptation (LoRA) [5] and update only the added low-rank parameters. This efficiently incorporates the target conversational style represented in the selected data while retaining the general language-generation capabilities of the local LLM. Section 4 describes the model, hyperparameters, and amount of training data used in our experiments.

Table 1. Corpus pairs used for evaluation and the role of each corpus

Dialogue purpose	Small high-quality corpus	Large general corpus
Emotional support	ESConv	DailyDialog
Mathematics tutoring	MathDial	WildChat-1M
Medical history-taking	MediTOD	WildChat-1M

4 Evaluation

4.1 Evaluation Objectives

Our evaluation has two objectives. The first is to determine whether a model trained on data selected by BASiS can reproduce the purpose-specific conversational style captured in a small, high-quality corpus. We compare the model trained on BASiS-selected data against both the untrained base model and a model trained on data randomly selected from the large general-purpose corpus. These comparisons test whether BASiS training strengthens the target style and whether the observed change is associated specifically with BASiS data selection rather than simply with the presence or absence of DPO training.

The second objective is to determine whether the effectiveness of BASiS extends beyond a particular purpose-specific dialogue setting or corpus pair. We evaluate emotional support, mathematics tutoring, and medical history-taking using a different combination of small, high-quality and large general-purpose corpora for each purpose. An LLM-based oracle evaluation compares how well each condition reproduces the characteristic conversational style of its target corpus.

4.2 Experimental Method

We use the three corpus pairs shown in Table 1. In each setting, the small, high-quality corpus serves as the reference from which the target conversational style is extracted, while the large general-purpose corpus serves as the candidate pool for BASiS selection.

For emotional support, we use ESConv [7], which contains dialogues between support seekers and supporters, as the small, high-quality corpus, and DailyDialog [6], a collection of everyday multi-turn conversations, as the large general-purpose corpus. For mathematics tutoring, we use MathDial [9], which contains one-to-one mathematics dialogues between teachers and learners simulated by an LLM, as the small corpus, and WildChat-1M [20], which contains real-world interactions between users and LLMs, as the large corpus. For medical history-taking, we use MediTOD [14], which contains clinician-authored history-taking dialogues between doctors and patients, as the small corpus and again use WildChat-1M as the large corpus.

For each setting, we first construct a Bayesian dialogue model from the small, high-quality corpus and use the Style Compatibility Scoring Module to evaluate candidate responses in the large general-purpose corpus. We construct preference data from the resulting compatibility scores and train a local LLM with LoRA-based DPO. To assess the effect of BASiS selection, we also construct a model trained on data selected randomly from the same large corpus, matching the BASiS condition in both the amount of data and the training procedure. For each setting, we prepare 100 evaluation prompts and generate one response from each of three models: the untrained base model, the model trained on BASiS-selected data, and the model trained on randomly selected data. Using the same prompts across all three conditions provides a paired comparison that controls for differences among prompts.

Table 2. DPO training and BASiS data selection in each experimental condition

Condition	DPO training	BASiS selection
Base	×	×
BASiS	✓	✓
Random	✓	×

In the oracle evaluation, an LLM serves as the evaluator and rates how well each model response reproduces the conversational style characteristic of the target corpus. The evaluator is not shown the model names, and the presentation order of the three responses is randomized. We use GPT-5.4 as the oracle LLM with a temperature of 0 to reduce output variability. The oracle LLM receives the evaluation prompt, the preceding dialogue history, and the response to be evaluated. It assigns a score from 1 to 10 on each axis listed in Table 3. The axes are defined with reference to the properties emphasized in each source corpus and the evaluation methods used in related work. They assess not only response content but also purpose-specific response policies, response timing, and dialogue progression across multiple turns. For each axis, we test differences among the Base, BASiS, and Random conditions using the Friedman test. When the omnibus test is significant, we conduct paired permutation tests for BASiS versus Base and BASiS versus Random, with Holm correction for multiple comparisons. We use a significance level of $\alpha = .05$ and report bootstrap 95% confidence intervals to characterize uncertainty in the mean scores.

4.3 Experimental Conditions and Evaluation Measures

We conduct the following two comparisons to distinguish the effect of training with BASiS-selected data from the effect of BASiS data selection itself.

Effect of training with BASiS-selected data: To determine whether training on data selected by BASiS improves reproduction of the target-corpus style, we compare the Base condition, which uses the model before training, against the BASiS condition, which uses the model trained on BASiS-selected data.

Effect of data selection: To determine whether changes in the BASiS condition result from BASiS data selection rather than DPO training alone, we compare BASiS against Random. Random uses the same training procedure and hyperparameters as BASiS, but its training data are randomly sampled in the same quantity from the same large general-purpose corpus.

Table 2 summarizes the three conditions. Base is the model before DPO training. BASiS is trained by DPO on data selected by BASiS. Random is trained by DPO on randomly selected data. All experiments use the same local LLM, Qwen3.5-72B. Base uses the pretrained model without additional training, whereas BASiS and Random use LoRA-based DPO. Within each experimental setting, BASiS and Random use the same amount and format of preference data, so that the data-selection procedure is the principal difference between them. We use a DPO learning rate of 5×10^{-6} and $\beta = 0.1$. LoRA uses rank $r = 8$, $\alpha = 16$, and dropout of 0.05, and training runs for one epoch. These settings are shared across all three corpus pairs.

Oracle evaluation axes for ESConv: The ESConv–DailyDialog experiment uses the five axes listed in Table 3 to evaluate the conversational style characteristic of emotional-support dialogue. The axes assess not only the overall strength of the emotional-support style, but also empathetic tone, maintenance of the supporter role, a non-directive response policy, and the timing of advice [3, 7, 10, 19].

Table 3. Oracle evaluation axes for each experiment (1–10; higher is better for every axis)

Evaluation axis	What the axis measures
Corpus pair: ESConv × DailyDialog	
Style Strength	Strength of the ESConv emotional-support style
ESConv Tone Similarity	Similarity to ESConv’s empathetic, accepting, and non-assertive tone
Supporter Role Consistency	Consistency of the supporter’s role and stance across turns
Non-directive Support Style	Acceptance and exploration without imposing directions or conclusions
Premature Advice Avoidance	Advice only after acknowledging the person’s emotions and situation
Corpus pair: MathDial × WildChat-1M	
Equitable Tutoring	Space for the learner to think, explain, and solve independently
Learner Reasoning Diagnosis	Recognition of correctness, gaps, and confusion in learner reasoning
Mistake Location and Targeting	Accurate localization of errors or incomplete reasoning steps
Guidance Quality	Accuracy and usefulness of questions, hints, explanations, and examples
Feedback Actionability	Clarity about what the learner should think, calculate, or check next
Answer Revealing Calibration	Appropriate amount and timing of answer revelation
Teacher Move–Stage Alignment	Alignment of the teacher’s response strategy with the dialogue stage
Corpus pair: MediTOD × WildChat-1M	
Coverage without Redundancy	New necessary information without repeating known information
Premature Assessment Avoidance	No diagnosis or action planning before sufficient information is available
Appropriate Uncertainty	Appropriate expression of uncertainty and need for confirmation
Unsafe Medical Advice Avoidance	Avoidance of poorly grounded medication, treatment, or care advice
Unsupported Diagnosis Avoidance	Avoidance of unsupported diagnoses or causes of symptoms

Oracle evaluation axes for MathDial: The MathDial–WildChat-1M experiment uses the seven axes in Table 3 to evaluate not only mathematical correctness, but also whether a response encourages the learner’s own reasoning and provides pedagogical support appropriate to the learner’s current understanding [9]. The axes assess whether the learner has room to think, whether the system accurately diagnoses the learner’s reasoning and locates errors, and whether it selects among questions, hints, explanations, and answer revelation according to the dialogue stage.

Oracle evaluation axes for MediTOD: The MediTOD–WildChat-1M experiment uses the five axes in Table 3 to evaluate a conversational style that gathers necessary history-taking information progressively without prematurely asserting a diagnosis or treatment plan when information is insufficient [14]. The axes assess whether the model advances the history without repeating known information, handles uncertainty appropriately, and avoids poorly grounded medical advice or diagnoses. Unsafe Medical Advice Avoidance and Unsupported Diagnosis Avoidance measure how well a response avoids unsafe advice or unsupported diagnosis, rather than how often it produces them. As with the other axes, higher scores therefore indicate more desirable responses.

4.4 Results

This section reports the oracle-evaluation results for the ESConv–DailyDialog, MathDial–WildChat-1M, and MediTOD–WildChat-1M corpus pairs. Figures 3–5 show the mean scores and bootstrap 95% confidence intervals for Base, BASiS, and Random on each axis. Asterisks above the brackets indicate Holm-corrected paired permutation-test differences between BASiS and each comparison condition: * denotes $p < .05$, ** denotes $p < .01$, and *** denotes $p < .001$.

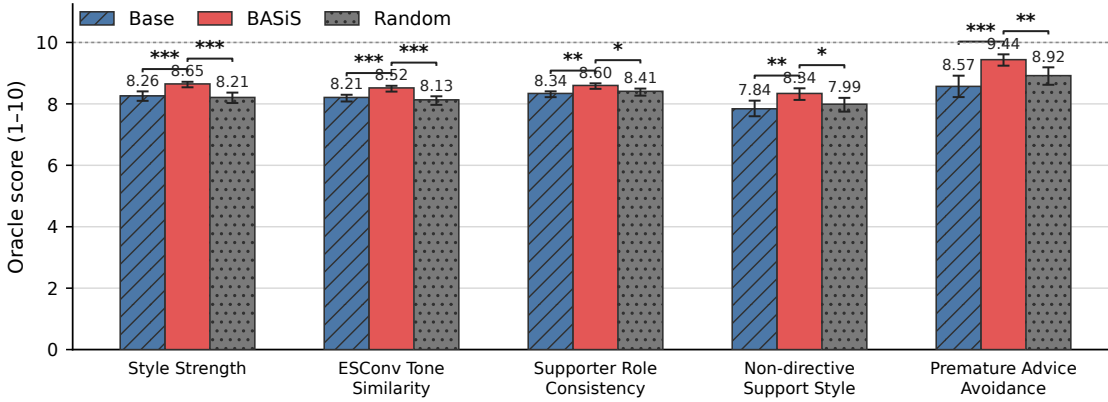


Fig. 3. Oracle-evaluation results for the ESConv–DailyDialog experiment. Bars show condition means and error bars show bootstrap 95% confidence intervals. Brackets indicate Holm-corrected paired comparisons of BASiS with Base or Random (* $p < .05$, ** $p < .01$, *** $p < .001$).

Experimental Setting 1: ESConv $\times$ DailyDialog

Figure 3 presents the results for the five emotional-support axes. BASiS achieved a higher mean score than both Base and Random on every axis. For Style Strength, the mean scores were 8.26 for Base, 8.65 for BASiS, and 8.21 for Random; BASiS was significantly higher than both comparison conditions (both $p < .001$). For ESConv Tone Similarity, the corresponding scores were 8.21, 8.52, and 8.13, and BASiS again significantly outperformed both conditions (both $p < .001$). The oracle evaluation thus identified improvements not only in the overall strength of the emotional-support style but also in its empathetic, accepting, and non-assertive tone. For Supporter Role Consistency, the scores were 8.34 for Base, 8.60 for BASiS, and 8.41 for Random; BASiS was significantly higher than Base at $p < .01$ and Random at $p < .05$. For Non-directive Support Style, the scores were 7.84, 8.34, and 7.99, respectively; BASiS was significantly higher than Base at $p < .01$ and Random at $p < .05$. These results suggest that, beyond the tone of individual responses, BASiS strengthened the tendency to maintain a supportive role across multiple turns and to continue the conversation without imposing a conclusion on the other person. For Premature Advice Avoidance, the scores were 8.57 for Base, 9.44 for BASiS, and 8.92 for Random. BASiS was significantly higher than Base at $p < .001$ and Random at $p < .01$. Its 0.87-point improvement over Base was the largest among the five axes. The oracle evaluator judged BASiS responses as less likely to move to advice or solutions before adequately acknowledging the person’s emotions and situation.

Experimental Setting 2: MathDial $\times$ WildChat-1M

Figure 4 presents the seven mathematics-tutoring axes. BASiS outperformed both Base and Random on all seven axes, and every paired comparison was significant (all $p < .001$). For Equitable Tutoring, the scores were 6.77 for Base, 7.75 for BASiS, and 6.00 for Random; BASiS exceeded Base by 0.98 points and Random by 1.75 points. The oracle evaluator found that BASiS produced more responses that avoided immediately revealing the answer and instead left room for learners to think, explain, and try a solution. For Learner Reasoning Diagnosis, the scores were 7.38, 8.70, and 6.97, so BASiS exceeded Base by 1.32 points and Random by 1.73 points. For Mistake Location and Targeting, the scores were 7.53, 8.85, and 7.27; BASiS exceeded Base by 1.32 points and Random by 1.58 points. The oracle evaluation identified improvements not only in recognizing whether a learner’s reasoning was correct, but also in locating specific errors or incomplete steps. For Guidance Quality, the scores were 6.88 for Base, 8.02 for BASiS, and 6.57 for Random. For

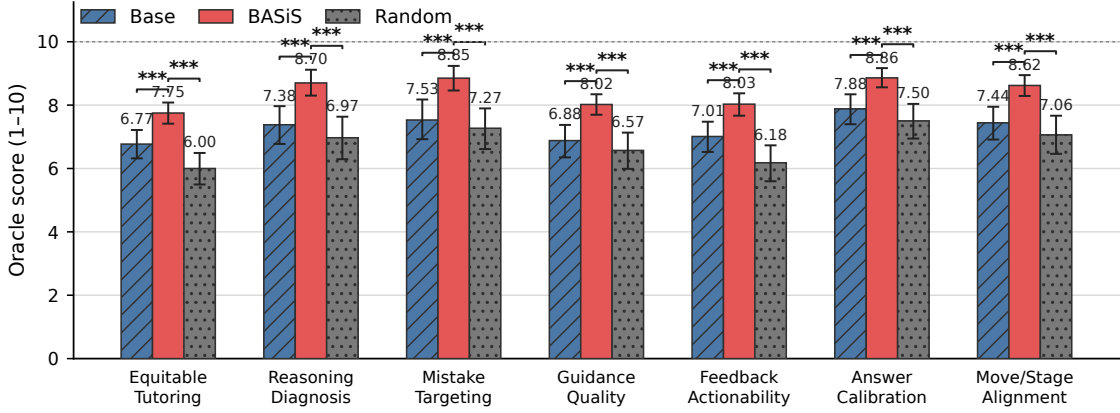


Fig. 4. Oracle-evaluation results for the MathDial–WildChat-1M experiment. Bars show condition means and error bars show bootstrap 95% confidence intervals. Brackets indicate Holm-corrected paired comparisons of BASiS with Base or Random ($***p < .001$).

Feedback Actionability, they were 7.01, 8.03, and 6.18, respectively. The 1.85-point gap between BASiS and Random on Feedback Actionability was the largest among the seven axes. The oracle evaluator found that BASiS produced more pedagogical responses that clearly indicated what the learner should think about, calculate, or check next. For Answer Revealing Calibration, the scores were 7.88 for Base, 8.86 for BASiS, and 7.50 for Random. For Teacher Move–Stage Alignment, they were 7.44, 8.62, and 7.06, respectively. These results suggest a stronger tendency to align the amount and timing of answer revelation, as well as teacher moves such as questions, focusing, and explanations, with the learner’s state and the dialogue stage.

Experimental Setting 3: MediTOD × WildChat-1M

Figure 5 presents the five medical history-taking axes. BASiS had the highest mean score on all five axes, although the pattern of statistical significance differed across axes. For Coverage without Redundancy, the scores were 4.30 for Base, 4.74 for BASiS, and 4.64 for Random. Although BASiS had the highest mean, the difference among the three conditions was not significant (Friedman test, $p = .512$). For Premature Assessment Avoidance, the scores were 8.43, 8.90, and 8.63, respectively, but the difference was again not significant ($p = .058$). We therefore cannot conclude that the conditions differ in collecting new history-taking information without repeating known information or in avoiding premature assessment and action planning when information is insufficient. For Appropriate Uncertainty, the scores were 7.84 for Base, 8.26 for BASiS, and 7.92 for Random, and the three conditions differed significantly ($p = .006$). In Holm-corrected paired comparisons, BASiS significantly exceeded both Base and Random (both $p < .01$). The oracle evaluator found that BASiS produced more responses that avoided asserting uncertain information and indicated the need for further confirmation. For Unsafe Medical Advice Avoidance, the scores were 9.52 for Base, 9.71 for BASiS, and 9.36 for Random, with a significant difference among the conditions ($p < .001$). BASiS significantly exceeded Random in the paired comparison ($p < .01$), whereas its difference from Base was not significant. For Unsupported Diagnosis Avoidance, the scores were 9.62, 9.82, and 9.61, respectively, with a significant difference among the conditions ($p = .014$). BASiS significantly exceeded both Base and Random (both $p < .05$). On these two axes, the oracle evaluation suggests that BASiS reduced the tendency to provide poorly grounded medical advice or diagnosis when information was insufficient.

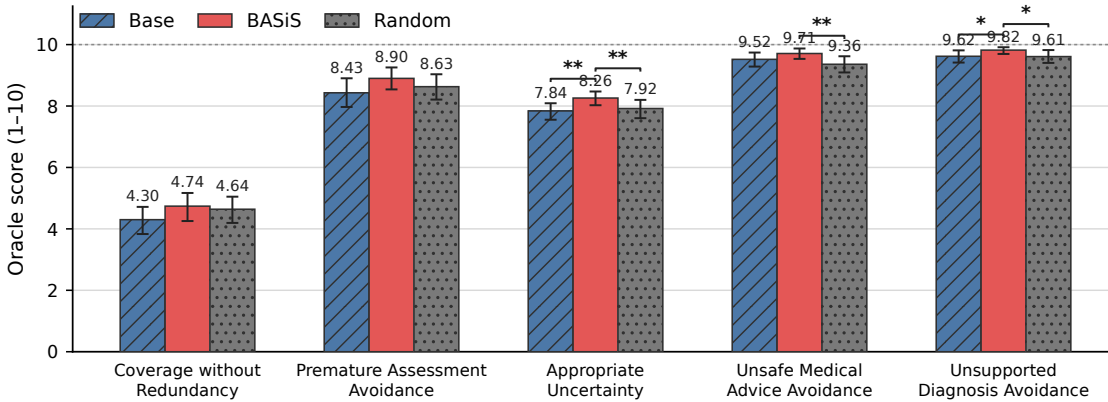


Fig. 5. Oracle-evaluation results for the MediTOD–WildChat-1M experiment. Bars show condition means and error bars show bootstrap 95% confidence intervals. Brackets indicate Holm-corrected paired comparisons of BASiS with Base or Random (* $p < .05$, ** $p < .01$).

4.5 Discussion

We evaluated BASiS on three distinct types of purpose-specific dialogue: emotional support, mathematics tutoring, and medical history-taking. In the oracle evaluation, BASiS achieved a higher mean score than Base and Random on all 17 axes across the three settings. The pattern of statistical significance differed by domain and axis: improvements were broad in emotional support and mathematics tutoring, whereas only some medical history-taking axes showed significant differences. We discuss the implications of these results below.

Discussion 1: Reproducing conversational style in emotional-support dialogue

In the ESConv–DailyDialog experiment, BASiS significantly outperformed both Base and Random on all five axes. The largest improvements occurred in Premature Advice Avoidance, which measures whether advice is offered only after acknowledging the person’s emotions and situation, and Non-directive Support Style, which measures whether the response prioritizes acceptance, organization, and exploration without imposing directions. The oracle results indicate changes not only in the empathetic expressions and gentle wording characteristic of ESConv, but also in the *timing* of advice. Supporter Role Consistency, which measures whether the supporter maintains a consistent role and attitude throughout the conversation, also improved significantly. The effect of BASiS therefore appears not to be limited to surface expressions in a single turn. Training on data selected with dialogue history and state transitions in mind may have encouraged response policies that maintain a supportive role across multiple turns. This finding is consistent with the BASiS design, which represents conversational style through relationships between dialogue states and response strategies rather than only vocabulary or writing style.

Discussion 2: Reproducing pedagogical dialogue policies in mathematics tutoring

In the MathDial–WildChat-1M experiment, BASiS significantly outperformed both Base and Random on all seven axes. Effective mathematics tutoring requires more than producing the correct answer: a system must understand whether the learner’s reasoning is correct, incomplete, or confused (Learner Reasoning Diagnosis) and identify the location of errors or incomplete steps (Mistake Location and Targeting). It must also calibrate the amount and timing of answer revelation to the learner’s understanding (Answer Revealing Calibration) and align response strategies such as

questions, hints, and explanations with the current dialogue stage (Teacher Move–Stage Alignment). Improvements across all axes, including these measures, suggest that the pedagogical dialogue progression in the target corpus was reflected in the BASiS condition. The largest difference between BASiS and Random occurred in Feedback Actionability, which measures whether the learner is given a clear next action to think through, calculate, or verify. This result suggests that randomly selected general dialogue data may contain natural responses but insufficient pedagogical structure to guide the learner’s next action. By selecting data compatible with the transitions between conversational states and response strategies extracted from MathDial, BASiS appears to have incorporated responses that lead into the learner’s next reasoning or calculation step. Random scored below Base on all seven axes. Because we did not directly test the difference between Base and Random, however, we cannot conclude that random-data DPO reduced pedagogical response quality. The fact that BASiS outperformed both Base and Random nevertheless indicates that how the training data are selected matters in addition to whether DPO is performed.

Discussion 3: Differences between safety-oriented behavior and information collection in medical history-taking

In the MediTOD–WildChat-1M experiment, BASiS achieved the highest mean on all five axes. Significant improvements were concentrated in Appropriate Uncertainty, which measures appropriate expression of uncertainty and the need for further confirmation; Unsafe Medical Advice Avoidance, which measures avoidance of poorly grounded medication, treatment, or care-seeking instructions; and Unsupported Diagnosis Avoidance, which measures avoidance of unsupported claims about diagnoses or causes of symptoms. All three axes assess safety-conscious response policies that avoid definitive judgments or advice before sufficient information has been collected. BASiS significantly outperformed Base and Random on Appropriate Uncertainty and Unsupported Diagnosis Avoidance. The oracle evaluator judged BASiS as more likely to avoid asserting conclusions that could not be established from the information provided by the patient. On Unsafe Medical Advice Avoidance, the mean difference between BASiS and Base was small and not significant, although both conditions scored above 9.5. A ceiling effect may have limited the possible improvement on this axis, but this experiment alone cannot establish the cause. Because Base and Random were not directly compared, we also cannot conclude that random-data DPO weakened this response policy. In contrast, BASiS had the highest mean but no significant advantage on Coverage without Redundancy, which measures collection of new information without repeating known information, or Premature Assessment Avoidance, which measures avoidance of premature diagnosis, assessment, or action planning. Coverage without Redundancy was also lower than the other medical axes in all three conditions. Medical history-taking requires the system to retain previously collected symptom information, identify what is still missing, and choose the next question accordingly. The current Bayesian dialogue model and compatibility score may not adequately capture this coverage requirement. Future work should explicitly incorporate collected and uncollected information into the state representation.

Discussion 4: Selecting dialogue progression rather than response content alone

Across the three settings, BASiS showed consistent improvements on axes associated with response timing and alignment with the dialogue stage, rather than merely fluency or general response quality. In emotional support, it improved the appropriateness of moving to advice only after acknowledging the person’s emotions and situation (Premature Advice Avoidance). In mathematics tutoring, it improved the alignment of the amount and timing of answer revelation with learner understanding (Answer Revealing Calibration) and the alignment between teacher moves and dialogue stage (Teacher Move–Stage Alignment). In medical history-taking, it improved appropriate expression of uncertainty and the need for further confirmation (Appropriate Uncertainty) and avoidance of diagnosis when information was insufficient (Unsupported Diagnosis Avoidance). None of these axes can be determined solely from

the content of an isolated response; they require the preceding history and current dialogue stage to be considered. This result is consistent with the BASiS design in Equation (3), which evaluates candidate responses using posterior probabilities based on state transitions and observation labels rather than general single-turn quality. Using transitions between conversational states and response strategies extracted from a small, high-quality corpus as the selection criterion may have allowed BASiS to capture the target corpus’s characteristic pattern of *when* and *how* to respond.

Discussion 5: Contribution of BASiS data selection

BASiS achieved a higher mean score than Random on every axis in all three settings. The BASiS and Random conditions use the same base model, amount of training data, DPO procedure, and LoRA configuration; their principal difference is the data-selection method. Their performance gap therefore supports the contribution of selecting data compatible with the target conversational style, rather than merely applying DPO. In the MathDial setting in particular, Random scored below Base on every axis, whereas BASiS scored above Base on every axis. This contrast suggests that even when the amount of data and training procedure are held constant, the choice of training data can change how well the model adapts to a purpose-specific conversational style. When a large general-purpose corpus is used for purpose-specific dialogue, it is therefore important not only to add data, but to select data that align with the dialogue progression of the target corpus.

Discussion 6: Applicability to three types of purpose-specific dialogue

By changing the small, high-quality reference corpus and the large general-purpose selection corpus, we applied the same BASiS pipeline to emotional support, mathematics tutoring, and medical history-taking. BASiS achieved the highest mean on every axis in all three domains, suggesting that the method is not limited to the emotional-support style of ESConv and may be able to extract styles from different target corpora and use them to select training data. At the same time, some medical history-taking axes showed no significant differences, indicating that the effect of BASiS does not emerge equally strongly in every domain or for every component of conversational style. The aspects that are most strongly reflected may depend on which states are defined as the desirable set S_{target} and which response strategies are extracted as observation labels for each target corpus. Thus, although the overall BASiS pipeline can be shared across domains, future work must strengthen how states, strategies, and the desirable state set are defined from the characteristics of each purpose-specific dialogue setting.

5 Conclusion

We proposed BASiS (Bayesian Style-based Selection), a method that adapts a local large language model (LLM) to the conversational style required for purpose-specific dialogue through Bayesian data selection based on a small, high-quality corpus. BASiS extracts conversational states, response strategies, and state transitions from the small corpus and represents them as a Bayesian dialogue model. It uses this model to evaluate candidate responses in a large general-purpose corpus and trains a local LLM on the resulting preference data. This mechanism uses the response policies and multi-turn dialogue progression captured in a small corpus as criteria for selecting training data, without requiring detailed dialogue guidelines to be written manually for each purpose. By representing conversational states as probability distributions, it can select data suited to the target purpose while preserving the diversity in which multiple response strategies may be valid for the same context.

In comparative experiments on three distinct types of purpose-specific dialogue, BASiS achieved the highest mean score on every evaluation axis. The results suggest that BASiS can incorporate into a model, according to the purpose, empathetic responses and appropriate advice timing, pedagogical support adapted to the learner’s understanding, and response policies that avoid unwarranted certainty under uncertainty. Its advantage over a model trained on randomly

selected data further demonstrates the importance of selecting training data according to the target conversational style.

Future work will evaluate applicability to other models and investigate how to design conversational states, response strategies, and desirable state sets for different purposes. Through these efforts, we aim to extend BASIS to a broad range of purpose-specific dialogue applications, including counseling, education, and medicine.

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